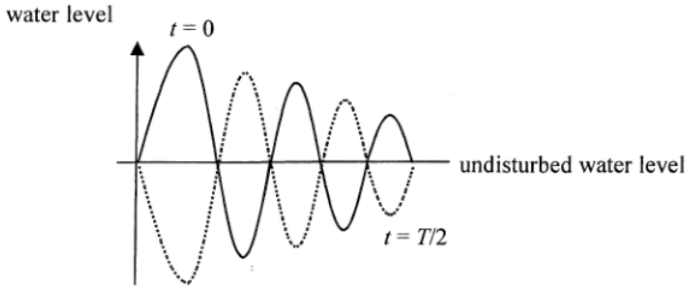
6. (b)



1A

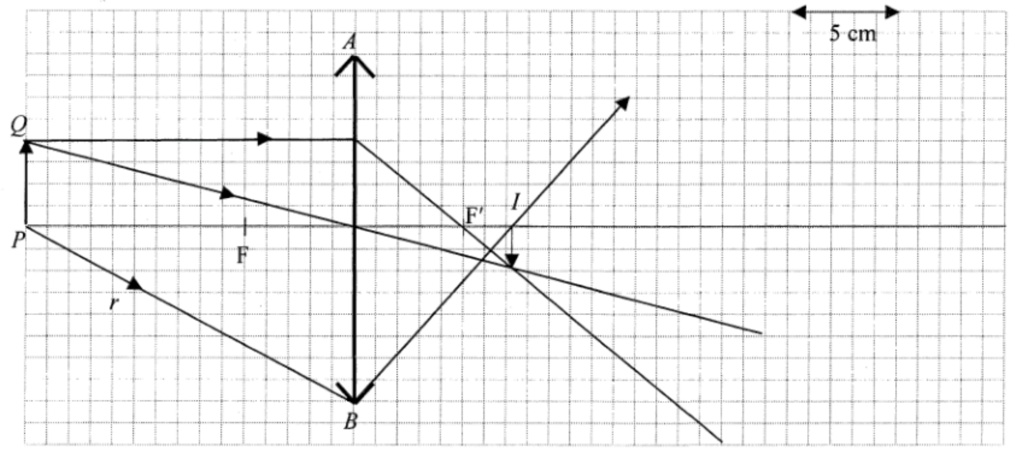
1

(c) The two waves are out of phase at Q as path difference = 3.5λ ($QS_1 = 4\lambda$ and $QS_2 = 7.5\lambda$), destructive interference occurs. 1M 1A 2

(d) $\Delta y = \dfrac{D\lambda}{a} = \dfrac{2.5 \times 550 \times 10^{-9}}{0.5 \times 10^{-3}}$ 1M

$= 2.75 \times 10^{-3}$ m 1A 2

7. (a) (i)



Two correct rays to find image I. 2A

Nature: real, inverted, diminished. 2A 4

(ii) Ray r correctly completed. 1A 1

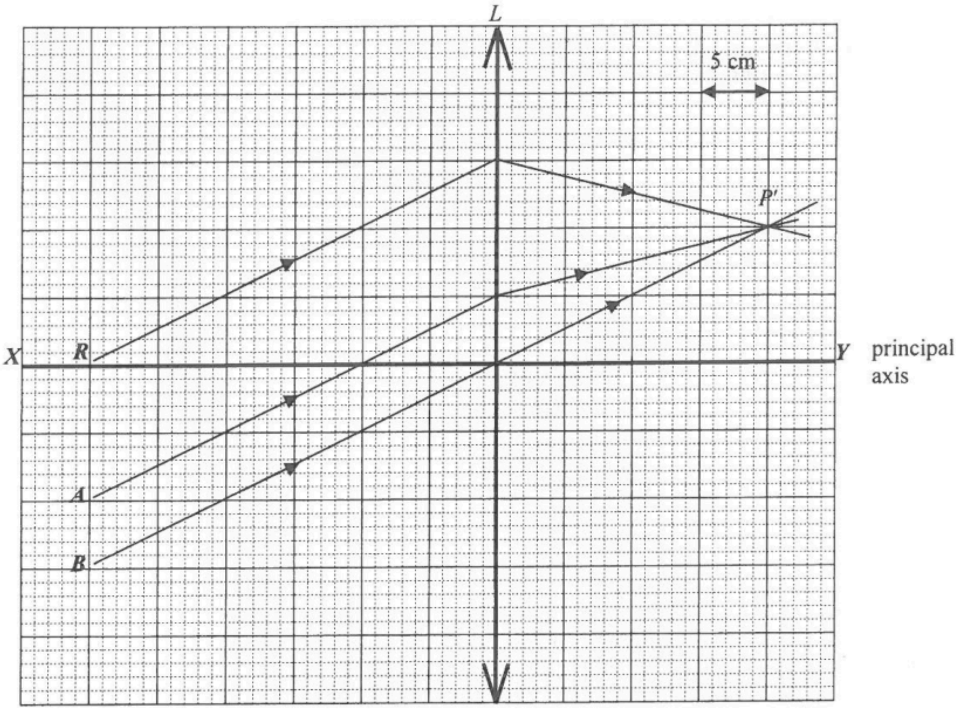
(b) (i) $\dfrac{1}{u} + \dfrac{1}{v} = \dfrac{1}{f}$ $\quad \dfrac{1}{15} + \dfrac{1}{v} = \dfrac{1}{10}$ 1M

$v = 30$ cm 1A

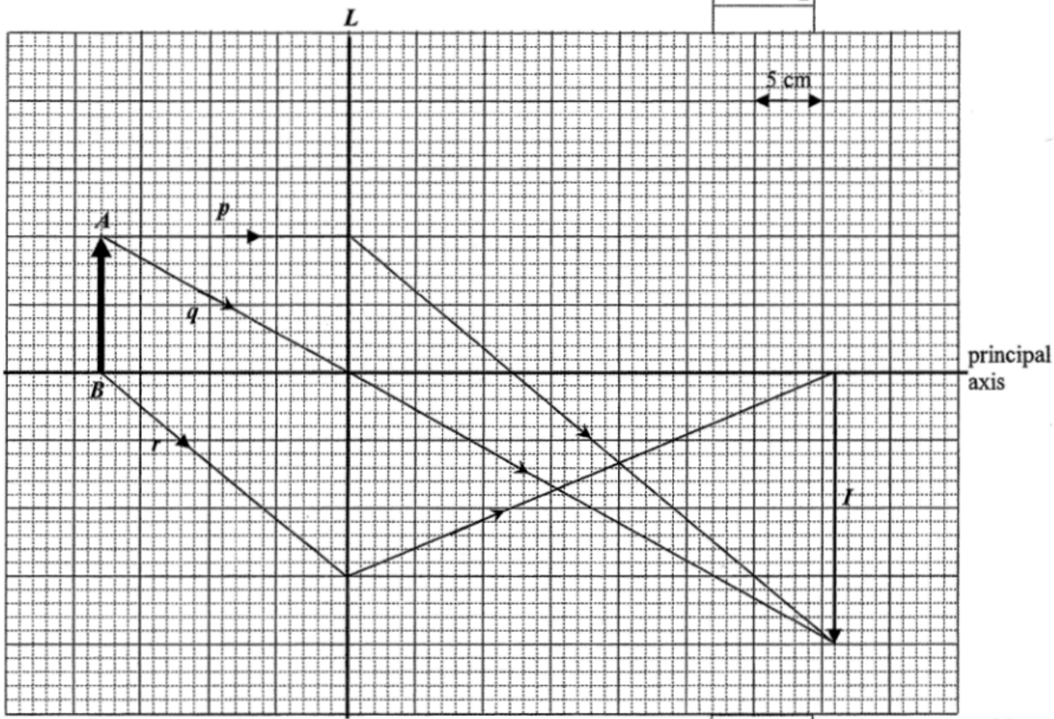
$m = \dfrac{30}{15} = 2$ 1A 3

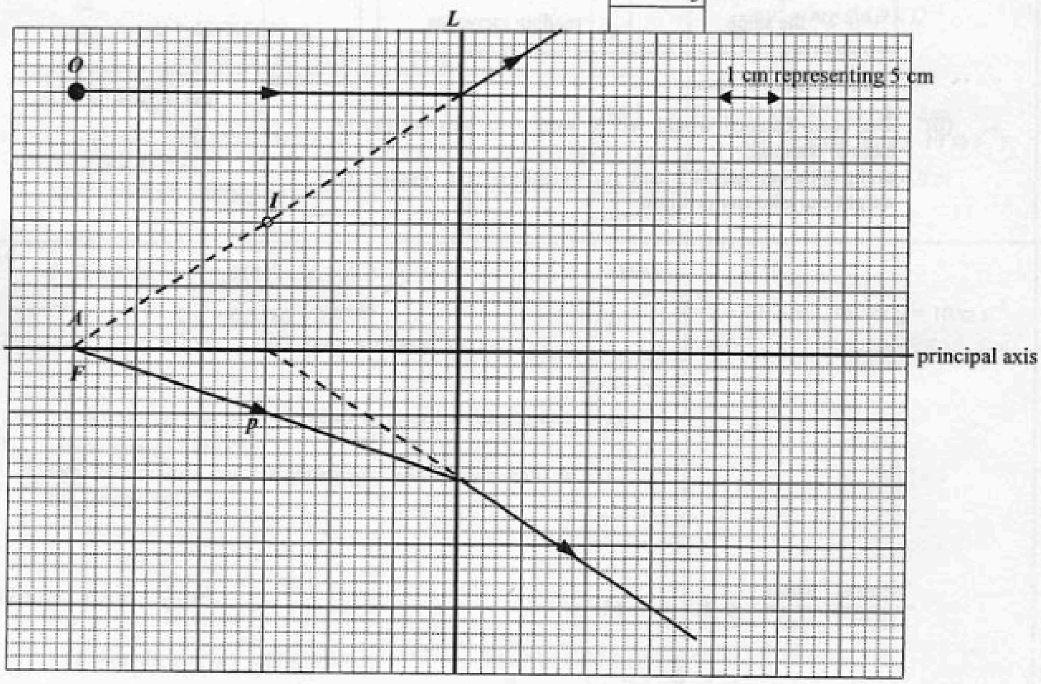
(ii) As the light energy collected by the lens is the same for both cases,
for (b)(i), an enlarged image ($u < v$),
same amount of light energy distributed over a larger image/intensity of light decreases as distance increases, 1A
i.e. dimmer image for (b)(i) 1A

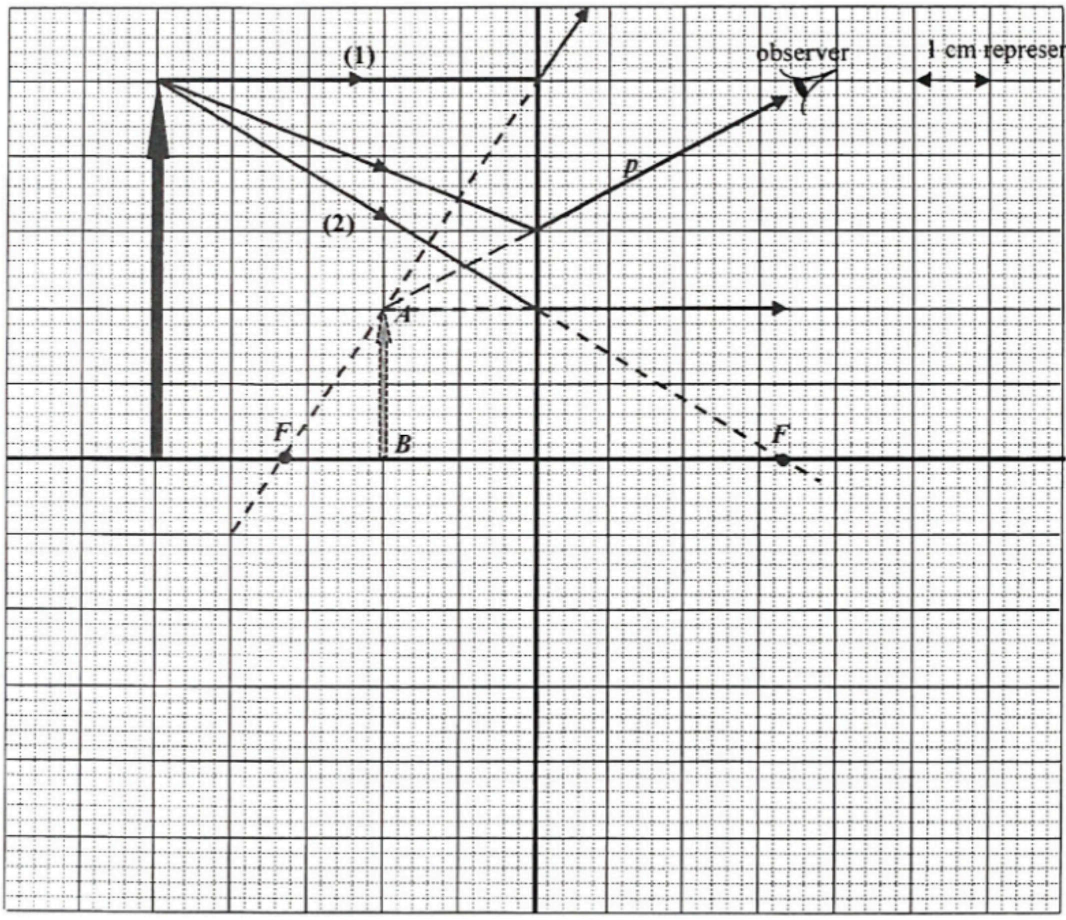
Or for (a), a diminished image ($u > v$),
same amount of light energy distributed over a diminished image,
i.e. brighter image for (a) 2

6. (a)	Convex lens / Converging lens.	1A		
	Refracted ray of *A* after passing through *L* bends towards the principal axis (or optical centre) / converges / bends inward / bends downward.	1A		
	<u>Or</u> A real / an inverted image can be formed.	1A		
	<u>Or</u> The object and the image are on opposite sides of the lens.	1A		
			2	
(b) (i)				
	Rays *A* and *B* correctly completed.	1M		
	P′ correctly located.	1M		
			2	
(ii)	$f = 20$ cm	1A		Accept 19 - 21 cm
			1	
(c)	Ray *R* correctly completed.	1M		
			1	
(d)	Use a screen to capture a/the (sharp) image (of a distant object).	1A		
	The distance between the screen and the lens is *f*.	1A		
			2	

6. (a)	- high temperature gradient; <u>or</u> - long path lengths for light rays.	1A	
	<u>Or</u> total internal reflection occurs.	1A	
		1	
(b) (i)	$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = n_4 \sin\theta_4$	1M	
	$\sin\theta_1 = \frac{n_4}{n_1}\sin\theta_4$		
	$\theta_1 = \sin^{-1}(\frac{1.000221}{1.000261})$	1M	
	$= 89.5°$ $(89.488°)$	1A	
		3	
(ii)	$\frac{h}{L} = \tan\alpha = \frac{1}{\tan\theta_1}$	1M	
	$L = h\tan\theta_1 = 1.5\tan 89.5° = 167.72$		
	$= 168\text{ m}$	1A	Accept 167.7 m to 172.0 m
	<u>Or</u> $L = \frac{h}{\tan\alpha} = \frac{1.5}{\tan 0.5°} = 171.88$	1M	
	$= 172\text{ m}$	1A	
		2	
(c)	The same distance away (168 m) because	1A	
	the illusion of 'water source' is caused by reflection of the light of distant objects at the same fixed angle.	1A	
	[i.e. $\alpha = 90° - 89.488° = 0.512°$ with the horizontal]		
	<u>Or</u> As long as the conditions for bending of the light and internal reflection are still the same,	1A	
	the 'water source' remains 168 m away (satisfying the same conditions / angle of reflection).	1A	
		2	

5.	(a)	(i)	Convex (lens) Only convex lens forms real image (which can be captured by a screen) Or Concave lens always forms virtual image (which cannot be captured by a screen). Or The image is formed on the other side of the lens.	1A 1A 1A 1A	
				2	
		(ii)	 opaque screen	1A	
				1	
	(b)	(i)	Image distance $v = 54 - 18 = 36$ cm ($D = 54$ cm) Magnification $= \frac{v}{u} = \frac{36}{18} = 2$	1A 1M/1A	
				2	
					
		(ii)	Image I of AB Rays p and q Ray r	1M 1M 1M	1M for correct position of I
				3	
		(iii)	Focal length = 12 ± 0.5 cm	1M/1A	
				1	
		(iv)	Move the lens 18 cm farther away from the object. Or move the lens 18 cm closer to / towards the screen. Height ratio = 1 : 4.	1M 1M 1A	
				2	

		Marks	Remarks
6. (a)	Place the (cylindrical) lens on the piece of paper and trace its outline, use ray tracing method e.g. - direct and trace a light ray towards the lens. - direct and trace a light ray along / parallel to the principal axis	1A	Or Using lens formula: - use ray tracing method: direct and trace a light ray (not parallel to the principal axis) towards the lens
	Direct another light ray parallel to the first one / principal axis (by shifting the ray box) and trace the path(s) of the (emerging) ray(s) on the paper.	1A	- extend (the path of) the emerging ray backward and locate the point of intersection (with the principal axis).
	Extend (the path of) the emerging rays (backward) and locate the point of intersection (on the focal plane containing F).	1A	- measure the corresponding object distance u and image distance v (on the principal axis)
	Measure the distance from the point of intersection (or F) to the centre of the lens, which gives the focal length of the lens.	1A	- sub. u and v into lens formula to find f
	Source of error: Scale uncertainty/accuracy/precision of the plastic ruler (read to nearest mm). OR Unable to mark the path correctly because of the thickness of the beam of light from the ray box OR The ray(s) is/are not parallel (to the principal axis) OR Any reasonable answer	1A	Accept: Using diagram(s) to supplement / simplify the description.
		5	
(b)			
(i)	L is diverging/concave. - only a diverging lens can produce a (virtual / erect / diminished) image between the object and the lens	1A 1A	
		2	
(ii)	Focal length = 30 cm Correct ray to find F	1A 1A	Accept (30 ± 1) cm
		2	
(iii)	Correct ray p	1A	
		1	

6.	(a)	L is diverging / concave. Only diverging / concave lens forms diminished, virtual image.	1A 1A 2	
	(b)			
		Correct position and height of object	2A 2	
	(c)	Correct ray to locate F and focus F correctly marked. Focal length = 16.5 cm	2M 1A 3	Accept: (15.5 ~ 17.5) cm
	(d)	Correct ray p from tip of object	2A 2	

6. (a) (i) $f = \frac{c}{\lambda}$

$= \frac{3\times10^8}{675\times10^{-9}}$ 1M

$= 4.444444 \times 10^{14}$ Hz $\approx 4.44 \times 10^{14}$ Hz 1A

2

(ii) $\frac{\sin 30°}{\sin\theta} = \frac{c}{v} = \frac{\lambda}{\lambda'}$

$= \frac{675}{450}$ 1M

$\sin\theta = \left(\frac{450}{675}\right)\sin 30°$

$\theta = 19.471° \approx 19.5°$ 1A

2

Definition: $n = \frac{\sin\theta_{air}}{\sin\theta'}$

(iii) The refractive index of glass for blue light is greater (than that for red light). 1A

1

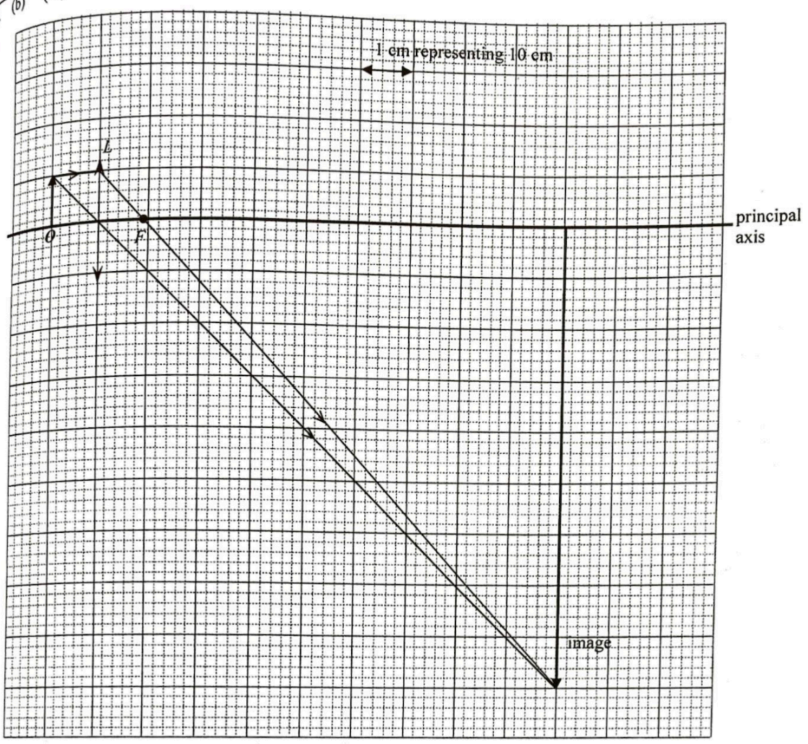
(b) (i) real and/or inverted 1A

1

(ii) 10 cm 1A

1

(b) (iii)



Location of F	1M	
Correct light rays from object to image	2M	
Correct position and size of image	1A	
Focal length of L = 9 cm	1A	Accept 8 cm to 10 cm
	5	

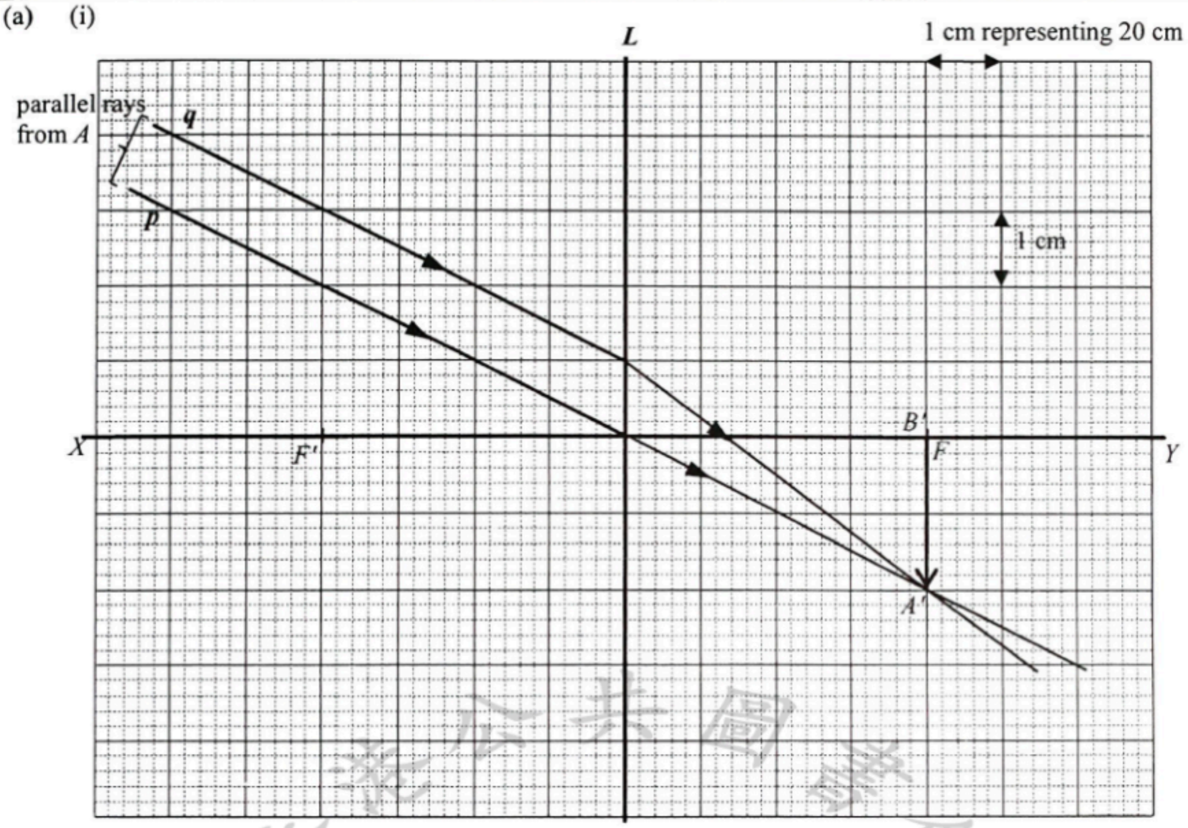
(iv) White light composes of different colours. 1A

Since the refractive indices of glass (the medium of the lens) are different for different colours,

the sizes of the colour images on the screen are slightly different.

When these images overlap on the screen, colour edges are seen. 1A

2

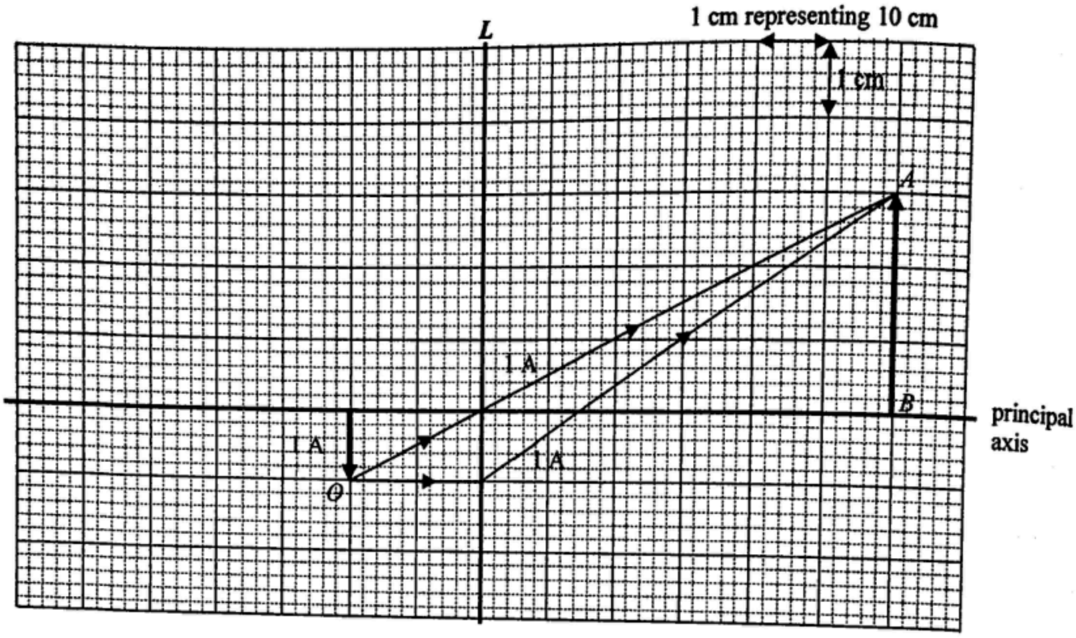
Solution	Marks	Remarks
5. (a) (i) 		
refracted rays of p, q correctly drawn and	1A	
A' correctly marked.	1A	
image $A'B'$ correctly marked.	1A	
	3	
(ii) Image $A'B'$ of a distant object, say, a building is real and therefore it can be captured	1A	
by a screen (placed at the focal plane).	1A	
	2	
(b) (i) Ratio by similar triangles, $\dfrac{\text{height of } AB}{\text{object distance}} = \dfrac{\text{height of } A'B'}{\text{focal length / image distance}}$		
$= \dfrac{2}{4 \times 20} = \dfrac{1}{40}$	1M	Accept height of image $A'B'$: 1.8 cm ~ 2.2 cm
$= 0.025$	1A	
	2	
(ii) $\dfrac{\text{height of } AB}{\text{object distance}} = \dfrac{1}{40} = 0.025$		
Height of $AB = 0.025 \times 200 = 5$ m	1A	
	1	

6. (a) Convex lens — 1A
because only a convex lens can form a real image. — 1A

Or
the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.

2

(b) (i)(ii)



3

(iii) Ray diagram method:
$f = 15$ cm — 1A

Accept: 14 cm ~ 16 cm

Or
By $\frac{v}{u} = 3$,
$\frac{0.6}{u} = 3$, i.e. $u = 0.2$ m
By $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$,
$\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$
$f = 0.15$ m or 15 cm

1

(c) Move / adjust the support / smartphone closer to lens *L*. — 1A

1

(d) The box should be painted black inside. — 1A

1